Human Ancestors



Intro

Effect of environment

Australopithecines

Introduction

- Interest in human origins since Darwin wrote *The Descent of Man* in 1871
  - Speculation about what evolutionary transition between ape and human may involve
  - Scientists began searching for fossils
  - Neanderthal fossils had already been discovered
  - Dubois looked in the tropics – found *Homo erectus* fossil
  - Since, many more fossils have been found
    - Lots of discoveries in 1970s onward in Africa.

Effect of environment on Hominin Evolution

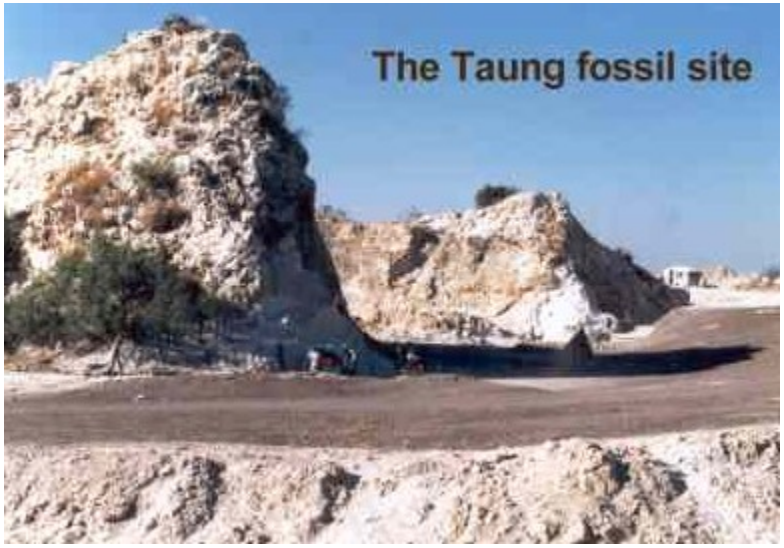
- Evolution by natural selection meant environment must have favoured certain traits
  - Hominins originally lived in woodland environment
    - Ape-like
    - Arms and hands suited to arboreal lifestyle
    - May have used some bipedal movement to get through forest clearings – like modern Orangutans.
  - Environmental change 5-6 million years ago
    - Temperatures fell
    - Forest areas diminished
    - Open grassland increased

Effect of environment on Hominin Evolution

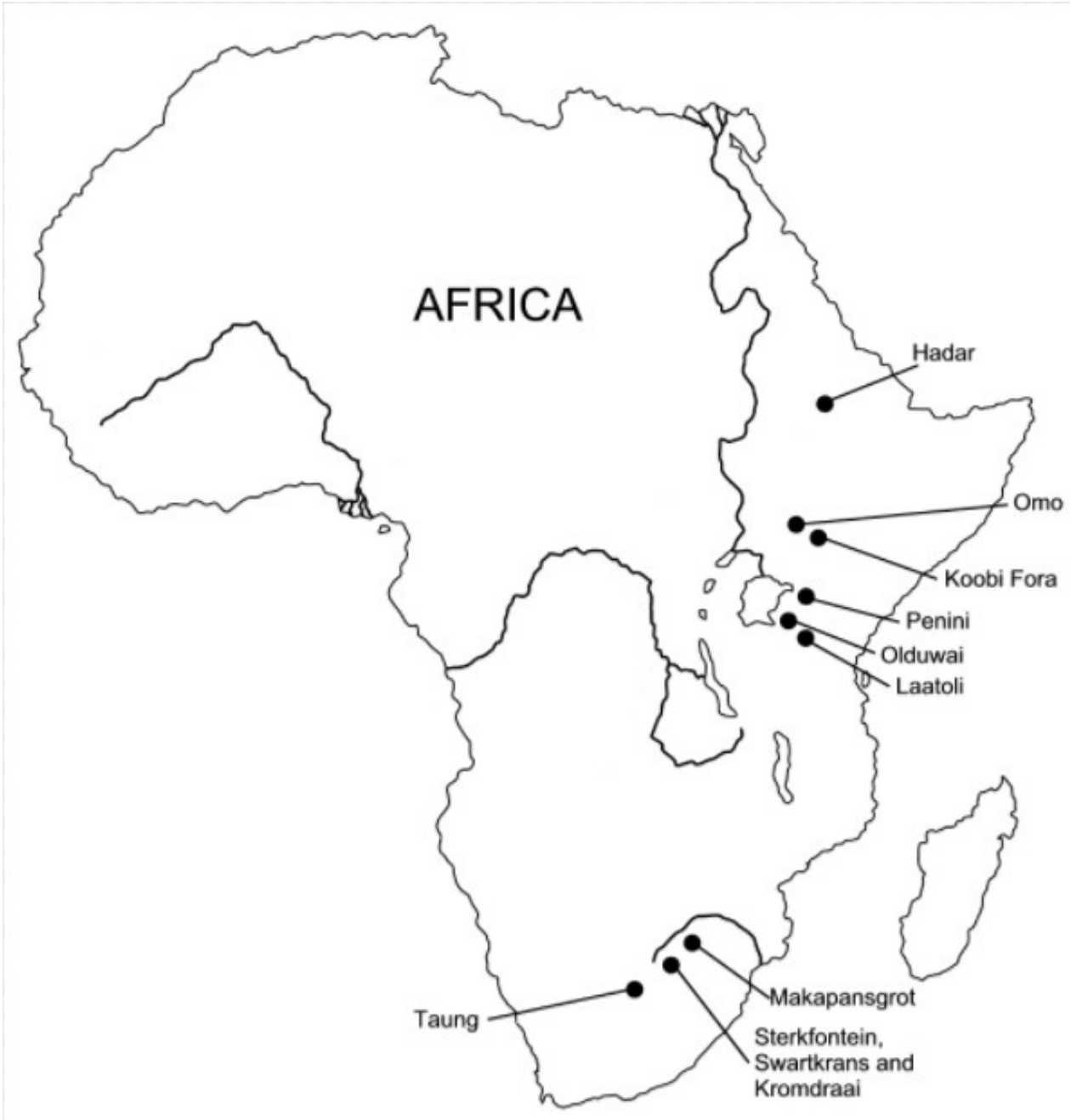
- Increase in open grassland caused bipedalism and other traits to be advantageous
  - Increased range of vision to detect prey and predators
  - Increased height/size deterred predators
  - Hands free for carrying food, using tools etc
  - Higher reach to pick fruit or collect honey
  - Improved body cooling
    - Temperatures fell
    - Forest areas diminished
    - Open grassland increased
- Food supply limited as forests replaced by grasslands
  - Evolutionary pressure
  - This was the time that Australopithecines were evolving

Australopithecines

- First fossil found in early 1920s
  - Raymond Dart (an Aussie!)
  - Found in the limeworks at Taung
  - Realised skull was not of an ape
  - Juvenile skull of a hominin
  - Need to be cautious about juvenile skulls - naturally more rounded
  - Jaws and teeth were also different
  - Foramen magnum positionining
  - Concluded that the skull was from an ancestral hominin
  - “Taung Child” – named *Australopithecus africanus*






AFRICA
Hadar
Omo
Koobi Fora
Penini
Olduwai
Laatoli
Taung
Makapansgrot
Sterkfontein,
Swartkrans and
Kromdraai

Australopithecines

- Further fossils found
  - By Robert Broom, at Sterkfontein, in 1930s
  - Different fossil found at Kromdraai – *Paranthropus robustus*
  - 1974: "Lucy" found in Hadar region of Ethiopia
    - 40% complete female skeleton
    - 3-3.6 million years old
    - *Australopithecus afarensis*
  - Laetoli footprints
    - 3 hominids walked across wet volcanic ash *Australopithecus afarensis*
    - Left behind footprints showing bipedal locomotion
    - 3.6 million years ago

Australopithecines

- Classification of *Australopithecus afarensis*

  Dentition

  - Dental arcade comparison
  - Size of canines, cusp prominence on molars

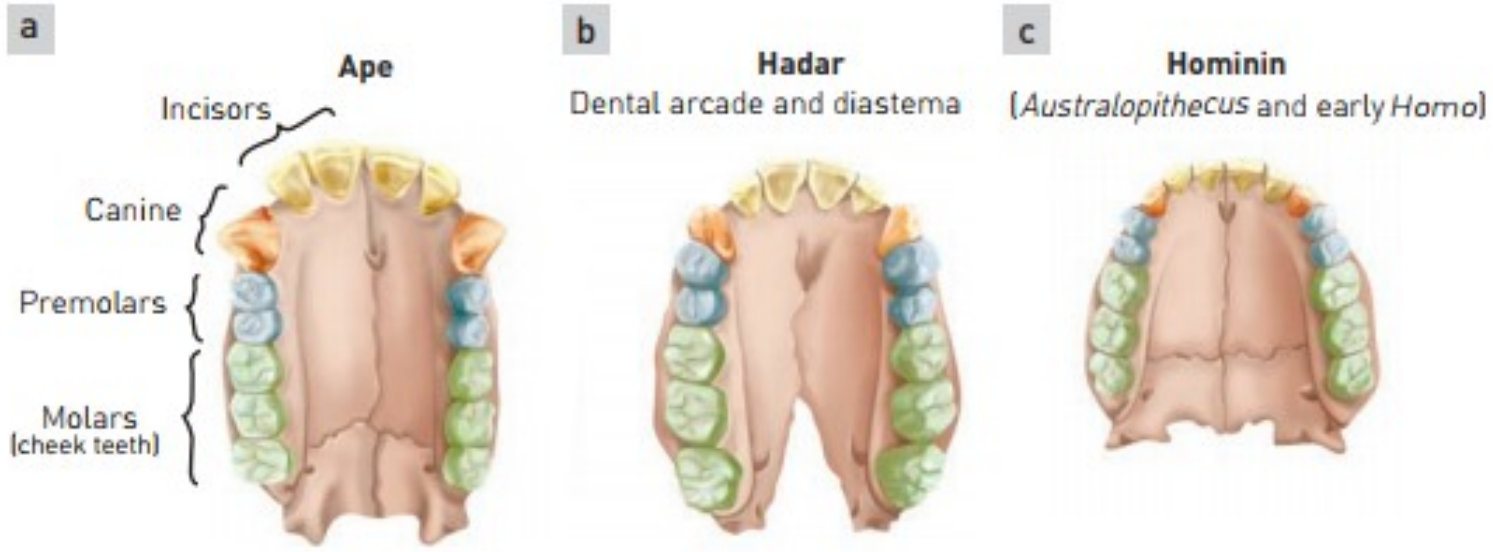


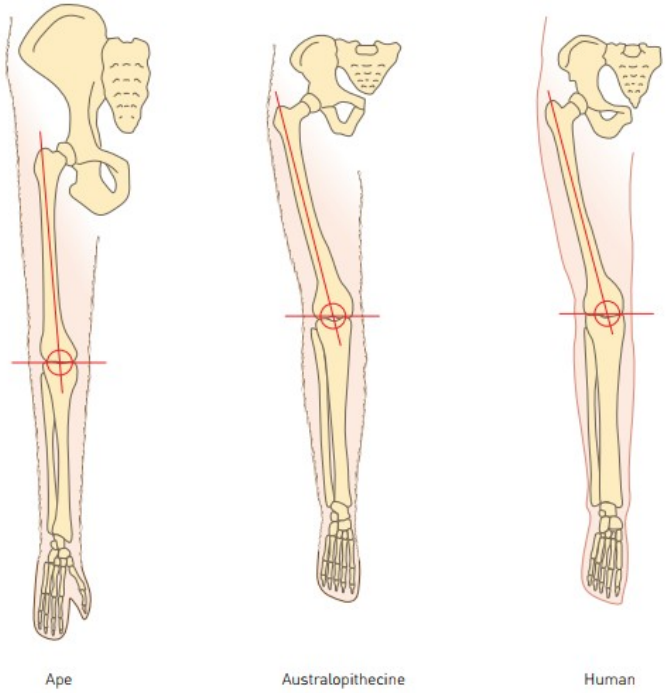
Figure 19.7 The dental arcades of **a** an ape, **b** a Hadar specimen and **c** an early hominin

Australopithecines

- Classification of *Australopithecus afarensis*

  Femur and Pelvis

  - Footprints show bipedal gait, although not quite the same as modern humans
  - Pelvis and carrying angle more similar to modern humans than to apes



Anatomical features of *Australopithecus*

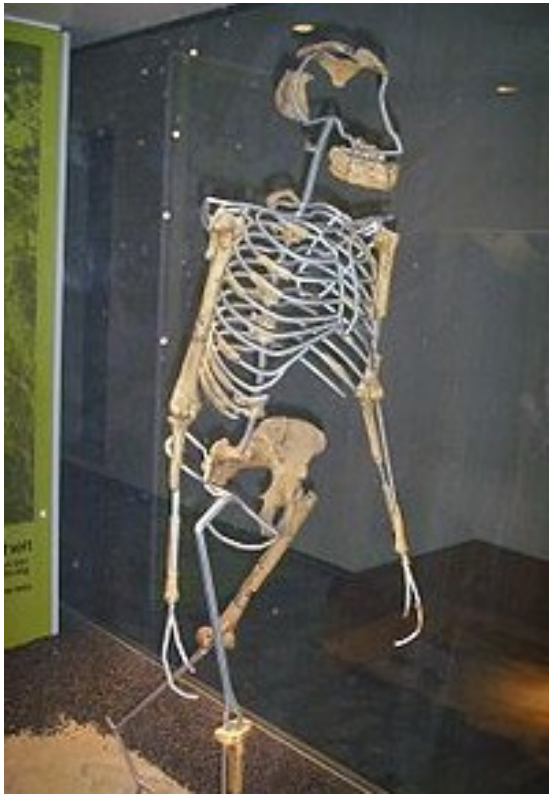
- Teeth typically hominin:
  - Canines short and non projecting
  - No gap between canines and premolars
  - Parabolic dental arcade
- Facial profile
  - Lower forehead, more projecting jaw than more modern hominins
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus smaller than gorilla
- Brain size and anatomy
  - Brain size 480cc, closer to modern gorilla than human
  - Relative brain size larger though: Australopithecus smaller than gorilla
  - Endocasts show brain more similar to humans in structure
  - Foramen magnum more forward than in apes
  - Skull more rounded in back than in apes

Anatomical features of *Australopithecus*

- Limb bones
  - Suggest bipedal locomotion
  - Pelvic and foot bones typically hominin
  - Non opposable, strongly built big toe
  - Thumb less mobile than modern humans, but more so than modern apes
  - Vertebral column – typically hominin curves
  - Foramen magnum – at base of skull.



Figure 19.9
A reconstruction of an australopithecine skull



Variation within the Australopithecines

- Probably several species within the genus *Australopithecus*
- Two main variants seen: gracile and robust
  - Gracile: finer boned, slender
  - Robust: stockier, heavier, more muscular

Gracile	Robust
120-140cm tall	Up to 170cm tall
25-35 kg	Up to 70 kg
Proportional molars	Larger, broader molars
Smaller jaw muscles	Large jaw muscles and attachments
Australopithecus africanus *Australopithecus afarensis*	Now classified into a different genus: *Paranthropus*. Eg: *Paranthropus robustus*



Robust form

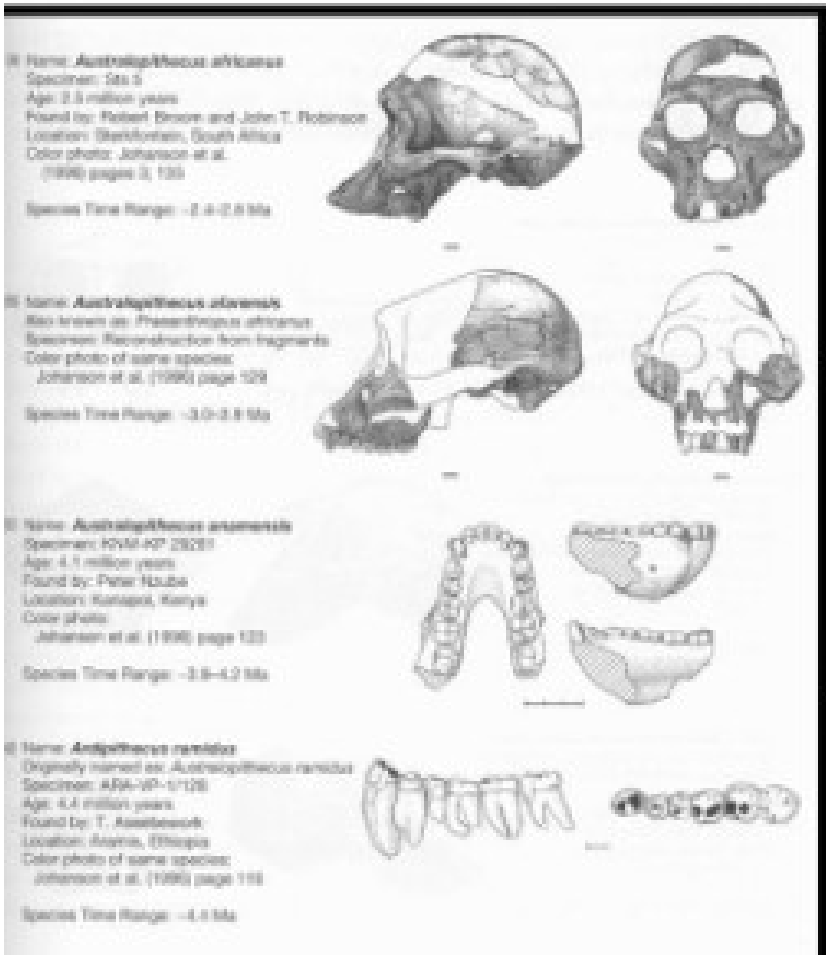


Gracile form

Gracile Australopithecines

Robust Australopithecines

Paranthropus



Name: ***Australopithecus africanus***
Specimen: Sts 5
Age: 2.5 million years
Found by: Robert Broom and John T. Robinson
Location: Sterkfontein, South Africa
Color photo: Johanson et al.
(1996) pages 3; 130

Species Time Range: ~2.4–2.6 Ma

Name: ***Australopithecus afarensis***
Also known as: *Praeanthropus africanus*
Specimen: Reconstruction from fragments
Color photo of same species:
Johanson et al. (1996) page 129

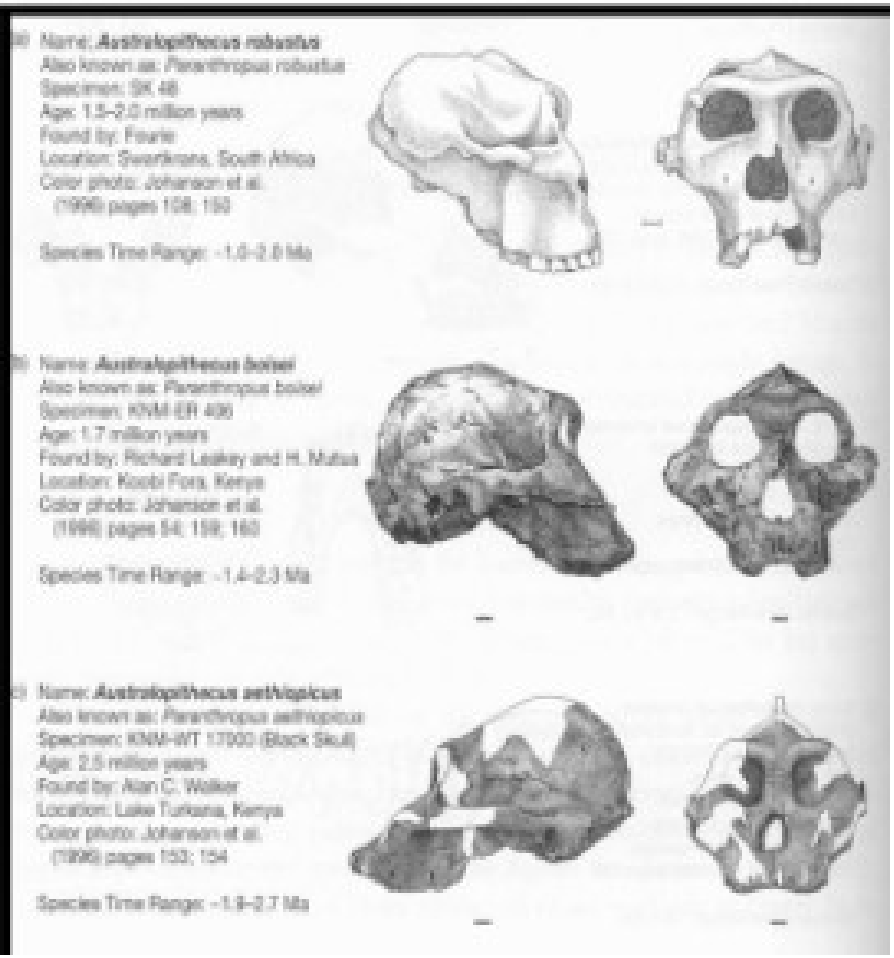
Species Time Range: ~3.0–3.8 Ma

Name: ***Australopithecus anamensis***
Specimen: KNM-KP 29281
Age: 4.1 million years
Found by: Peter Nzube
Location: Kanapoi, Kenya
Color photo:
Johanson et al. (1996) page 133

Species Time Range: ~3.9–4.2 Ma

Name: ***Ardipithecus ramidus***
Originally named as: *Australopithecus ramidus*
Specimen: ARA-VP-1/129
Age: 4.4 million years
Found by: T. Assefaw
Location: Aramis, Ethiopia
Color photo of same species:
Johanson et al. (1996) page 116

Species Time Range: ~4.4 Ma



Name: ***Australopithecus robustus***
Also known as: *Paranthropus robustus*
Specimen: SK 48
Age: 1.5–2.0 million years
Found by: Fourie
Location: Swartkrans, South Africa
Color photo: Johanson et al.
(1996) pages 108; 150

Species Time Range: ~1.0–2.0 Ma

Name: ***Australopithecus boisei***
Also known as: *Paranthropus boisei*
Specimen: KNM-ER 406
Age: 1.7 million years
Found by: Richard Leakey and H. Mutua
Location: Koobi Fora, Kenya
Color photo: Johanson et al.
(1996) pages 94; 159; 160

Species Time Range: ~1.4–2.3 Ma

Name: ***Australopithecus aethiopicus***
Also known as: *Paranthropus aethiopicus*
Specimen: KNM-WT 17000 (Black Skull)
Age: 2.5 million years
Found by: Alan C. Walker
Location: Lake Turkana, Kenya
Color photo: Johanson et al.
(1996) pages 153; 154

Species Time Range: ~1.9–2.7 Ma



